

Impulsivity, Reward and Addiction

Alcohol (ADH1B, ALDH2, GABRA2), nicotine (CHRNA5), dopamine and opioids (DRD2/ANKK1, COMT, OPRM1), cannabis (FAAH), addition scores, alcohol, tobacco and cannabis, and load of risk taking loci

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	7
6. Favourable	8
7. Unfavourable	9
8. Mechanisms of ageing	10
9. Practical suggestions	11
10. Limitations	12
11. Methods and sources	13
12. Annex: all variants analysed	14

1. Summary

Favourable. The common genetic factor of the additions is low (percentile 20.1). Alcohol use disorder is typical-low (percentile 35.4; problematic use in 56.1) and the load of alcohol loci is very low (7.5). In tobacco, the perturbation of use is in percentile 36.3 and the loci load is low (10.3); in §61, "having smoked" is in the 2nd percentile. There is neither the **DRD2 Taq1A A1** (G/G) nor the **GABRA2** risk allele.

Unfavourable. The beverage score per week is in percentile 94.6: if you drink, the tendency would be to drink more, but this does not count as long as alcohol is zero. Already having consumed cannabis is in the percentiles 89.3 and 91.2 (track linked to opening and looking for novelty), and the FAAH C385A is heterozygous. CHRNA5 D398N is heterozygous (more cigarettes per day in smokers; cigarette score per day in 70.5). The load of risk taking loci is in 83.7, in line with "considers someone who risks" in the 91st percentile (§69); the score of risk taking in 4 domains was in the 13th percentile. It does not have the **ADH1B His** protector. Out of this report, there are genetic ADHD (94 and 97 percentiles, §70), depression (the 95th percentile, §67) and insomnia (the 89th percentile loci, §70).

What does it mean? The general vulnerability to addiction is low and zero alcohol and the absence of tobacco are well supported by genetics. The point to be monitored is the search for novelty and reward (ADHD high genetics, taking risk, cannabis). This demand should be channeled to healthy rewards (training, projects, learning) and far from substances and rapid digital rewards.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 8 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 7 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 9 PGS Catalog scores calculated locally with 7.008.262 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 7189 loci, 6760 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more tendency (addiction, alcohol, tobacco, cannabis). The beverage score per week measures the amount of consumption in those who drink, not dependence.

Addition (common factor)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Common genetic factor of additions	PGS003849 (Hatoum 2023, PRS-CS, 585 thousand variants)	577.909 584.753 (weighted coverage 98%)	20,1	TIPIC	typical

Alcohol

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Problematic use of alcohol	PGS005214 (Deng 2024, lassosum, 454 thousand variants)	427.566 453.914 (weighted coverage 94%)	56,1	TIPIC	typical
Alcohol use disorder	PGS002738 (Lai 2022, PRS-CS, 326 thousand variants)	320.539 326.000 (weighted coverage 98%)	35,4	TIPIC	typical
Beverages per week	PGS003376 (Saunders 2022, LDpred, 1.06 million variants)	1.026.074 1.055.988 (weighted coverage 96%)	94,6	MAIS PROVABLE	unfavourable

Tobacco

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Start smoking	PGS003360 (Saunders 2022, LDpred, 1.06 million variants)	1.026.000 1.055.915 (weighted coverage 96%)	56,9	TIPIC	typical
Cigarettes per day	PGS003368 (Saunders 2022, LDpred, 1.06 million variants)	1.025.896 1.055.811 (weighted coverage 96%)	70,5	TIPIC	typical
Tobacco use disorder	PGS002037 (Privé 2022, LDpred2, 848 thousand variants)	831.077 847.691 (weighted coverage 97%)	36,3	TIPIC	typical

Cannabis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Have already consumed cannabis	PGS002125 (Privé 2022, LDpred2, 817 thousand variants)	800.574 816.604 (weighted coverage 97%)	89,3	MAIS PROVABLE	unfavourable
Have already consumed cannabis	PGS003504 (Ding 2023, LDpred2-auto, 980 thousand variants)	972.627 979.739 (weighted coverage 99%)	91,2	MAIS PROVABLE	unfavourable

4. Key variants by topic

Variants of known effect on alcohol and nicotine metabolism and dopamine, opioids and endocannabinoids systems (see also the §21 of dopamine and the §61 of caffeine).

Alcohol

ADH1B transforms alcohol into acetaldehyde and ALDH2 eliminates acetaldehyde. The fast variants of ADH1B and slow variants of ALDH2 give redness and protect from dependence. GABRA2 is a subunit of the GABA-A receptor, where alcohol acts.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ADH1B	rs1229984	R48H	C/C	imputation DR2=0.88	T (0 copy/s)	neutral	Arg48His: the T allele oxidizes ethanol very quickly (redness, malaise) and protects from alcohol dependence	Strong
ALDH2	rs671	E504K	G/G	chip	A (0 copy/s)	neutral	Glu504Lys: aldehyde dehydrogenase inactive (intensity flushing; protects from dependence)	Strong
GABRA2	rs279858		T/T	chip	G (0 copy/s)	favourable	GABA-A α 2 receiver: the G allele is associated with alcohol dependence and impulsivity (mixed evidence)	Weak

Nicotine

The nicotinic receptor α 5 (CHRNA5) regulates aversion to high doses of nicotine; the allele A reduces this "broil".

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CHRNA5	rs16969968	D398N	G/A	chip	A (1 copy/s)	unfavourable (light)	Asp398Asn: Allele A increases nicotine dependence and the number of cigarettes (detail in §61)	Strong

Dopamine, opioids and endocannabinoids

Dopamine (DRD2, COMT), endogenous opioids (OPRM1) and endocannabinoids (FAAH) form the circuit of reward and motivation (§21).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
DRD2/ ANKK1	rs1800497	Taq1A	G/G	chip	A (0 copy/s)	favourable	Taq1A: Allele A (A1) is associated with less D2 receptors in the striatum and more search for reward (mixed evidence; §21)	Weak
COMT	rs4680	V158M	G/G	chip	A (0 copy/s)	without judgment	Val158Met: Val/Val (G/G) = lower prefrontal dopamine ("guerreiro"; more resilience to stress, less working memory at rest)	Moderate
OPRM1	rs1799971	A118G	A/G	chip	G (1 copy/s)	without judgment	A118G: the G allele is associated with more reward with alcohol and the best response to naltrexone in alcohol dependence (mixed evidence)	Weak
FAAH	rs324420	P129T	C/A	chip	A (1 copy/s)	without judgment	C385A: the allele A low to FAAH (more anandamide); associated with less anxiety and more risk of problematic cannabis use in some studies	Weak

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of taking risk and adventure, tobacco (start smoking, cigarettes per day, dependence), alcohol (drinks per week, problematic use, disturbance of use) and disturbances of cannabis use, opioids and substances. Alleles are oriented towards more consumption or risk and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Risk taking and adventure	273 (266)	282 de 532	266	83,7	without judgment
Start smoking and tobacco	4718 (4441)	4378 de 8882	4441	10,3	favourable
Alcohol: consumption and problematic use	1446 (1329)	1365 de 2658	1328	7,5	favourable
Cannabis and opioids	752 (724)	719 de 1448	724	50,7	typical

6. Favourable

- **Common genetic factor of additions** in percentile 20.1 (Hatoum 2023: shared between alcohol, tobacco, cannabis and opioids). **Moderate.**
- **Alcohol:** perturbation of use in percentile 35.4, problematic use in the 56.1, loci load in the 7.5. **Moderate.**
- **Tobacco:** perturbation of the use in percentile 36.3, loci load in the 10.3; "I have smoked" in the 2nd percentile in §61 (start smoking in 56.9, a score from another study). **Moderate.**
- **DRD2/ANKK1 Taq1A G/G** (without A1): normal density of D2 receptors (§21). **Weak.**
- **GABRA2 rs279858 T/T:** without the allele associated with alcohol dependence. **Weak.**
- **Disorders of cannabis, opioids and substances use:** typical loci load (50.7). **Weak moderated.**

7. Unfavourable

- **Beverages per week** in percentile 94.6: tendency to a higher consumption if drinking. With zero alcohol, it has no effect. **Moderate.**
- **Have already consumed cannabis** in the 89.3 and 91.2 percentiles (shared trait with openness to experience and ADHD); **FAAH C385A C/A** (associated with problematic cannabis use in some studies). **Weak moderated.**
- **CHRNA5 D398N G/A**: 1 allele of nicotine dependence and more cigarettes per day (cigarettes per day in percentile 70.5; \$61 and \$75). Strong for the genotype, only relevant with nicotine.
- **Risk taking**: loci load in 83.7 ("it is considered someone who risks" in the 91st percentile, \$69; the 4 domain score was in the 13th percentile). Without judgment: it is a trait, not a disease. **Weak moderated.**
- **ADH1B C/C and ALDH2 G/G**: metabolism of normal alcohol, without the "broil" of redness that protects from dependence. Strong for the genotype (most people are like this).
- **Background (other reports)**: ADHD genetics in the percentiles 94 and 97 (\$70), depression in 95 (\$67) and insomnia in the loci in 89 (\$70). The three increase the search for self-medication with substances. **Moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Intercellular Communication	Signaling of dopamine (COMT Val/Val, DRD2 without A1), opioids (OPRM1 A/G) and endocannabinoids (FAAH C/A)	Mixed
Neuroinflammation	Alcohol, nicotine and smoked cannabis activate microglia; with zero consumption, the risk is low	Favourable (conditional)
Oxidative stress	Alcohol acetaldehyde and tobacco smoke are major sources of free radicals; avoided with zero alcohol and without tobacco	Favourable (conditional)
Sleep disturbance	Nicotine, alcohol and cannabis fragmented sleep; insomnia in the loci in the 89th percentile (§70)	Unfavourable (conditional)
Chronic Inflammation	Tobacco is the greatest preventable trigger of chronic inflammation; tobacco use disorder in percentile 36.3	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Keep the alcohol at zero.** With the beverage score per week in the 95th percentile and without the protective ADH1B, drinking "only socially" would tend to scale. *Suggestion (not applied).*
2. **Zero nicotine in any form** (cigarette, vape, bags, heated tobacco). CHRNA5 A/G and alfa-1 antitrypsin PiMZ (§75) add up. *Suggestion (not applied).*
3. **Avoid cannabis**, mainly smoked. The consumption scores are high, the genetic ADHD is high and insomnia, memory and mood (depression in the 95th percentile) worsen with regular use. *Suggestion (not applied).*
4. **Channeling the search for reward and novelty** for training, adventure sport with security, projects and learning. Setting limits to fast digital rewards (social networks, games, bets): maximum daily time and mobile phone outside the room. *Suggestion (not applied).*
5. **Pre-commitment** in risk decisions (financial, betting, driving): written rules and a "waiting period" of 24 hours before large decisions. *Suggestion (not applied).*
6. **Protecting sleep and mood** (insomnia and genetic depression): regular sleep, melatonin 0.3–1 mg of the plan, caffeine $\leq$ 200 mg only in the morning and daily exercise. If mood decreases or insomnia worsens, seek help early rather than self-medication. *Suggestion (not applied).*
7. **Opioids:** avoid prolonged use (load of loci for prolonged use of high opioids in the Pain and Analgesia report). LDN 4.5 mg already partially blocks the μ receptor. *Suggestion (not applied).*
8. **Information:** OPRM1 A/G is associated, in some studies, with the best response to naltrexone in alcohol dependence. The class GLP-1 (retactured) reduces the willingness to drink in trials. *Suggestion (not applied).*

10. Limitations

- Consumption scores ("ever consumed cannabis", "start smoking", "drinking per week") measure behaviors influenced by personality and by the environment, not dependence.
- The "ever smoked" of §61 (the 2nd percentile) and the "beginning to smoke" of this report (the 57th percentile) come from studies and different methods; the load of tobacco loci (low) and the disturbance of use (typical-low) support a favourable global reading.
- DRD2 Taq1A, GABRA2, OPRM1 and FAAH have small effects and inconsistent replication.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** GRCh38 files from PGS Catalog (9 scores: common factor of addictions, problematic use and alcohol use disturbance, drinks per week, start smoking, cigarettes per day, tobacco use disorder and two cannabis consumption). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Hatoum 2023 for the common factor of additions; Saunders 2022 for tobacco and alcohol; Zhou 2020 for the problematic use of alcohol Edenberg 2019 for ADH1B and ALDH2; Thorgeirsson 2008 for CHRNA5; Karlsson Linnér 2019 for risk taking; Johnson 2020 for cannabis, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Alcohol	ADH1B	rs1229984	C/C	imputation DR2=0.88	T	0	strand +	without the protective allele (common genotype)
Alcohol	ALDH2	rs671	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Alcohol	GABRA2	rs279858	T/T	chip	G	0	complement (strand)	without the risk allele
Nicotine	CHRNA5	rs16969968	G/A	chip	A	1	strand +	1 copy of risk allele
Dopamine, opioids and endocannabinoids	DRD2/ ANKK1	rs1800497	G/G	chip	A	0	strand +	without the risk allele
Dopamine, opioids and endocannabinoids	COMT	rs4680	G/G	chip	A	0	strand +	0 copy of the effect allele
Dopamine, opioids and endocannabinoids	OPRM1	rs1799971	A/G	chip	G	1	strand +	1 copy of effect allele
Dopamine, opioids and endocannabinoids	FAAH	rs324420	C/A	chip	A	1	strand +	1 copy of effect allele